Piecewise-Defined & Implicit Functions

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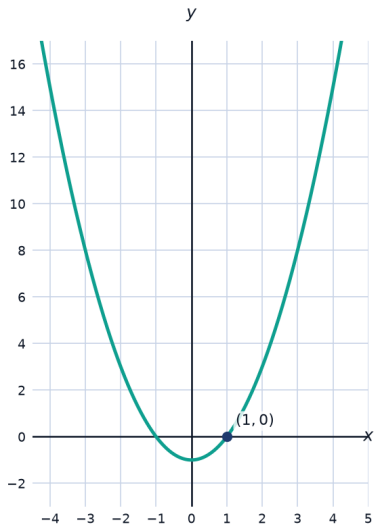
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Evaluating piecewise functions**

A shipping company charges a flat 5 dollars for packages up to 2 kg, plus 1.50 dollars per kg for anything over 2 kg. Write a piecewise function $C(w)$ for the cost, and find $C(5)$.

2 **[Foundational]** • **Target: Checking continuity of piecewise functions**

Find the value of k that makes $f(x) = 3x + k$ for $x < 2$, and $f(x) = x^2 - 1$ for $x \geq 2$, continuous at $x = 2$.



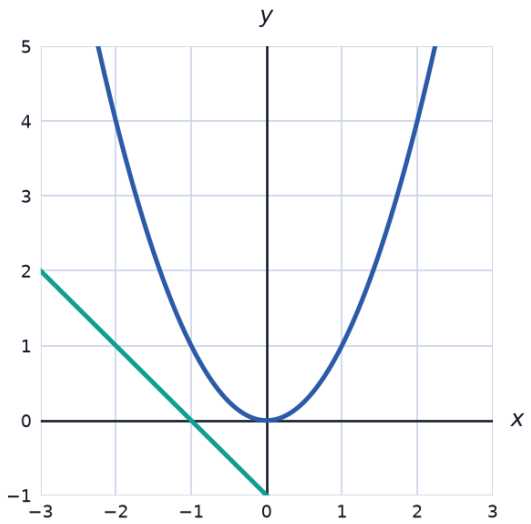
3 **[Foundational]** • **Target: Checking continuity of piecewise functions**

Determine whether $g(x) = x + 1$ for $x \leq 1$, and $g(x) = 3x - 2$ for $x > 1$, is continuous at $x = 1$, showing your reasoning.

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4 [Foundational] • Target: Visualizing a piecewise graph

The graph shows $f(x) = -x - 1$ for $x < 0$, and $f(x) = x^2$ for $x \geq 0$.

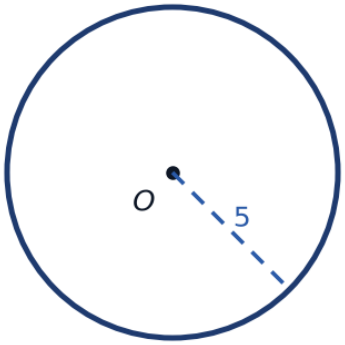


a Evaluate the left piece and the right piece of f at $x = 0$, and use this to determine whether f is continuous there.

b State the range of f shown on the graph.

5 [Foundational] • Target: Implicit relations and testing for functions

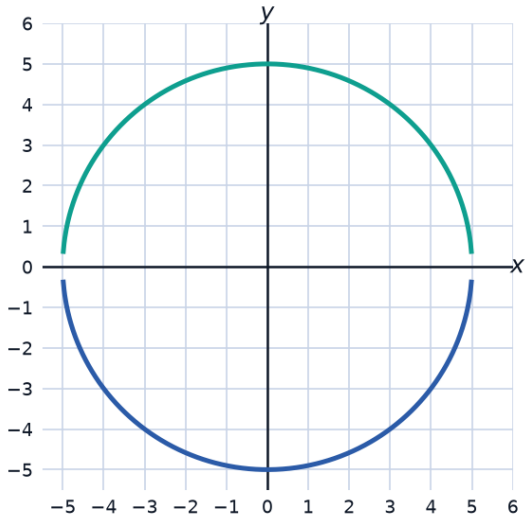
Determine whether $x^2 + y^2 = 25$ defines y as a function of x. Justify using the vertical line test.



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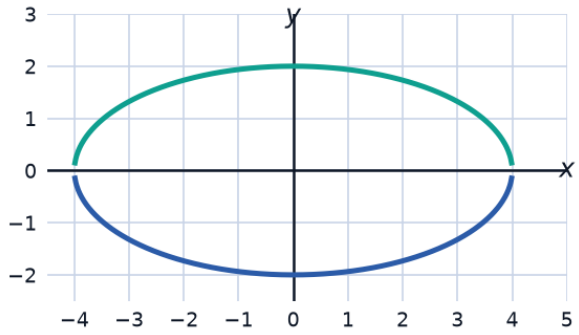
6 [Foundational] • Target: Implicit relations and testing for functions

Solve $x^2 + y^2 = 25$ for y explicitly, and state which branch corresponds to a function.



7 [Foundational] • Target: Implicit differentiation-style reasoning (algebraic slope)

For the implicit relation $x^2 + 4y^2 = 16$ (an ellipse), find the two y-values when $x = 2$, and describe what this means about the relation as a function.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Modeling with piecewise functions

A tax schedule charges 10% on income up to 10{,}000 dollars and 20% on income above 10{,}000 dollars. Write a piecewise tax function $T(x)$ for income x, and find the tax owed on an income of 25{,}000 dollars.

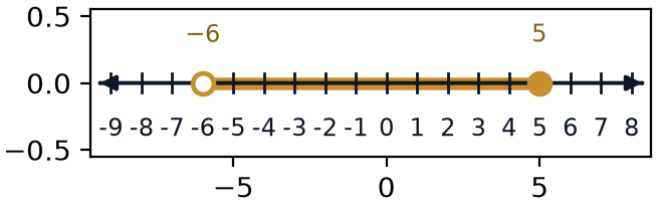
9 [Intermediate] • Target: Piecewise functions with three or more pieces

A parking garage charges 3 dollars for the first hour, 2 dollars for each additional hour up to 5 hours total, and a flat 15 dollars for anything over 5 hours. Write this as a three-piece function $P(h)$ for h hours parked, and find $P(4)$.

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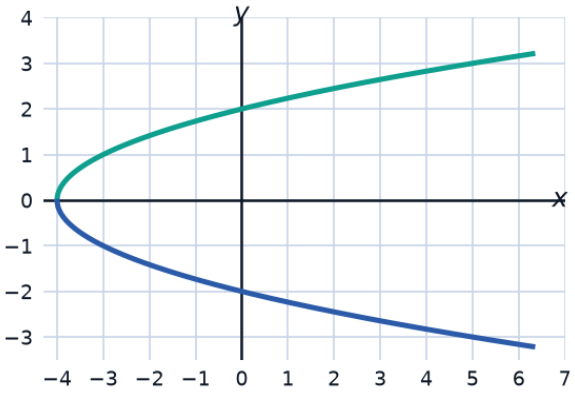
10 [Intermediate] • Target: Piecewise functions with three or more pieces

Evaluate the absolute value function $f(x) = |2x - 6|$ at $x = 1$ and $x = 5$ by first rewriting it as a two-piece function without absolute value bars.



11 [Intermediate] • Target: Graphing implicit relations point-by-point

For the implicit relation $y^2 = x + 4$, find all y-values when $x = 0$, $x = 5$, and $x = -4$, and state the smallest allowed x-value.



12 [Intermediate] • Target: Piecewise functions and average rate of change

For $f(x) = x^2$ on $x < 2$ and $f(x) = 3x - 2$ on $x \geq 2$, find the average rate of change of f on $[0, 4]$.

13 [Intermediate] • Target: Piecewise functions and average rate of change

Given $g(x) = 2x + 1$ for $x \leq 3$ and $g(x) = x^2 - 5$ for $x > 3$, find the average rate of change of g on $[1, 5]$.

14 [Intermediate] • Target: Piecewise functions and average rate of change

A cell phone plan charges 20 dollars flat for up to 2 GB of data, plus 8 dollars per GB for each additional GB, up to a maximum monthly charge of 68 dollars (unlimited data cap). Write a three-piece function $M(x)$ for the monthly cost based on x GB used, and find $M(8)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Modeling with piecewise functions • Cross-Lesson: 2.02 Logarithmic Functions & Properties

If $\sin\theta = -\frac{4}{5}$ and θ lies in Quadrant III, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.

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16 [Challenging] • Target: Piecewise functions with three or more pieces • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Let $f(x) = \frac{x^2 - 9}{x^2 - 5x + 6}$. Identify the locations of all vertical asymptotes and removable discontinuities (holes) of $f(x)$.

17 [Challenging] • Target: Graphing implicit relations point-by-point • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Find all real and complex roots of the polynomial $P(x) = 2x^3 + 3x^2 - 8x + 3$.

18 [Challenging] • Target: Piecewise functions and average rate of change • Cross-Lesson: 4.01 Inverse & Composite Functions

Evaluate the limit $\lim_{x \to 3} \frac{x^2 - 9}{x - 3}$ algebraically, and interpret what happens to the graph of $f(x) = \frac{x^2 - 9}{x - 3}$ at $x = 3$.

19 [Challenging] • Target: Evaluating piecewise functions • Cross-Lesson: 4.03 Parametric Equations & Modeling

Prove the trigonometric identity $\frac{1 - \cos 2\theta}{\sin 2\theta} = \tan \theta$ for all values of θ where both sides are defined.

20 [Challenging] • Target: Checking continuity of piecewise functions • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Solve the exponential equation $4^{x+1} - 5(2^x) + 1 = 0$ for all real values of x.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Evaluating piecewise functions

For the implicit relation $x^2 + y^2 = 625$, find all y-values when $x = 24$.

a. $y = \pm 25$

b. $y = \pm 24$

c. $y = \pm 7$

d. $y = 7$

22 [Foundational] • Target: Checking continuity of piecewise functions

A shipping company charges a flat 7 dollars for packages up to 3 kg, plus $\frac{5}{2}$ dollars per kg for anything over 3 kg. Find the cost for a 13 kg package.

a. 32

b. 33

c. 31

d. $\frac{79}{2}$

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23 [Foundational] • Target: Visualizing a piecewise graph

The function f is defined by $f(x) = \frac{2x^2 + kx - 6}{x - 2}$ for all real numbers $xeq2$, where k is a constant. If the graph of $y = f(x)$ in the standard (x, y) coordinate plane has a removable discontinuity (a hole) at the point $(2, L)$, what is the value of L?

a. -1

b. 3

c. 7

d. 11

24 [Foundational] • Target: Implicit relations and testing for functions

The graph of the sinusoidal function $y = A\sin(Bx + C) + D$ is shown below. The function has a maximum value of 5 and a minimum value of -1. The graph passes through the point $(0, 2)$ and has a period of $\frac{\pi}{2}$. Find the value of A and B. Assume $A > 0$ and $B > 0$.

a. $A = 3, B = 4$

b. $A = 3, B = 2$

c. $A = 6, B = 4$

d. $A = 6, B = 2$

25 [Foundational] • Target: Implicit differentiation-style reasoning (algebraic slope)

A piecewise function is defined by $f(x) = 4x + 2$ for $x < -3$, and $f(x) = 3x - 4$ for $x \geq -3$. Find $f(-4)$.

a. -15

b. -17

c. -14

d. -13

26 [Foundational] • Target: Modeling with piecewise functions

A piecewise function is defined by $f(x) = 2x + 3$ for $x < 1$ and $f(x) = x + k$ for $x \geq 1$. Find k so that f is continuous at $x = 1$.

a. 5

b. 6

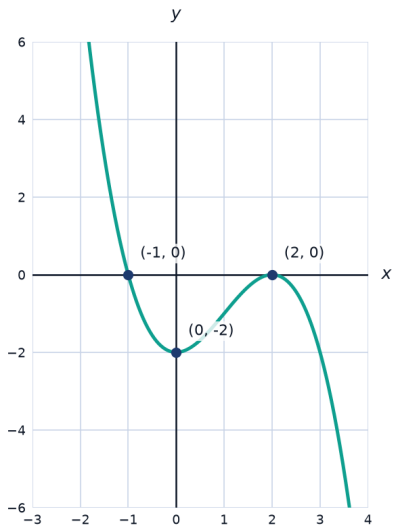
c. 4

d. 1

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27 [Foundational] • Target: Piecewise functions with three or more pieces

The graph of a third-degree polynomial function $P(x)$ is shown in the standard (x, y) coordinate plane below. The graph crosses the x-axis at $(-1, 0)$, is tangent to the x-axis at $(2, 0)$, and passes through the point $(0, -2)$. What is the value of $P(-2)$?



a. -16

b. -8

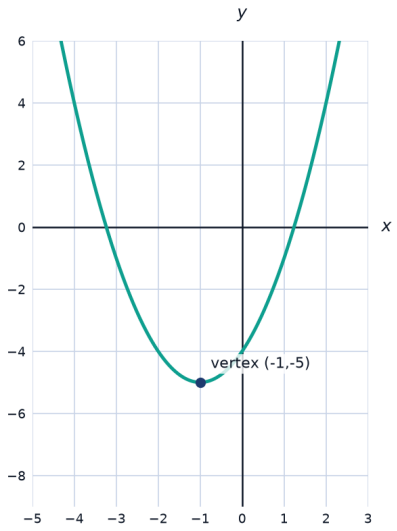
c. 4

d. 8

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28 **[Foundational]** • **Target: Graphing implicit relations point-by-point**

The graph of $g(x) = f(x + 1) - 5$ is obtained from $f(x)$ by:



a. Shifting 1 units left and 5 units down

b. Shifting 1 units right and 5 units down

c. Shifting 1 units right and 5 units up

d. Reflecting over the x-axis

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Piecewise functions and average rate of change**

Consider a point $P(x, y)$ that lies on the parabola defined by the equation $y = x^2 - 6x + 13$. Let F be the focus of this parabola. If the distance from P to the directrix of the parabola is minimized, what is the y-coordinate of the point P?

a. 3

b. 4

c. 5

d. 6

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30 [Intermediate] • Target: Evaluating piecewise functions

What is the sum of all real values of k for which the equation $|x^2 - 6x + 5| = 2x + k$ has exactly three distinct real solutions?

a. -3

b. -1

c. -2

d. -13

31 [Intermediate] • Target: Checking continuity of piecewise functions

The quadratic function $f(x) = ax^2 + bx + c$ has its vertex at $(-1, 5)$. If the graph of $f(x)$ also passes through the point $(2, -13)$, find the value of b.

a. 3

b. -3

c. 6

d. -6

32 [Intermediate] • Target: Visualizing a piecewise graph

Consider the function $f(x) = ax^2 + bx + c$. The graph of $y = f(x)$ is a parabola that passes through the points $(-1, 5)$ and $(2, -7)$. If the vertex of the parabola is located at $x = -1/2$, find the value of c.

a. 1

b. 2

c. 3

d. 4

33 [Intermediate] • Target: Implicit relations and testing for functions

Consider the rational function $f(x) = \frac{ax^2 + bx + c}{dx^2 + ex + f}$, where a, b, c, d, e, f are non-zero constants. Given that $\lim_{x \to \infty} f(x) = \frac{1}{2}$ and $\lim_{x \to -1} f(x) = \infty$, determine the value of $\frac{a}{d}$.

a. $\frac{1}{2}$

b. 1

c. 2

d. Cannot be determined

34 [Intermediate] • Target: Implicit differentiation-style reasoning (algebraic slope)

In the xy-plane, an angle θ is in standard position. If the terminal side of θ passes through the point $(-5, 12)$, what is the value of $\sin(\theta)$?

a. $\frac{12}{13}$

b. $-\frac{5}{13}$

c. $-\frac{12}{13}$

d. $-\frac{12}{5}$

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35 [Intermediate] • Target: Modeling with piecewise functions

Consider the function $f(x) = ax^2 + bx - 5$. If the graph of $f(x)$ passes through the points $(-2, 3)$ and $(1, -3)$, find the value of $a - b$.

a. 1

b. -1

c. 2

d. -2

36 [Intermediate] • Target: Piecewise functions with three or more pieces

The function f is defined by $f(x) = -x^2 + bx + c$, where b and c are constants. In the standard (x, y) coordinate plane, the maximum value of $f(x)$ is 14, and $f(0) = 5$. If $b > 0$, what is the value of $f(5)$?

a. 5

b. 10

c. 13

d. 14

37 [Intermediate] • Target: Graphing implicit relations point-by-point

Consider the rational function $f(x) = \frac{x^2 - 9}{x - 3}$. What is the value of $f(x)$ as x approaches 3?

a. 0

b. 6

c. Undefined

d. 3

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Piecewise functions and average rate of change • Cross-Lesson: 4.01 Inverse & Composite Functions

Consider the rational function $f(x) = \frac{x^2 - 4}{x - 2}$. What is the value of the function as x approaches 2?

a. 4

b. 0

c. Undefined

d. 2

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39 [Challenging] • Target: Evaluating piecewise functions • Cross-Lesson: 4.03 Parametric Equations & Modeling

The functions f and g are defined for all real numbers by $f(x) = 2x^2 - 3$ and $g(x) = 3x + 1$ What is the value of $f(g(-2))$?

a. -53

b. 16

c. 47

d. 95

40 [Challenging] • Target: Checking continuity of piecewise functions • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

For $0 < x < 1$, simplify $\cos\left(\sin^{-1}(x)\right)$ as an algebraic expression in x.

a. $\sqrt{1 - x^2}$

b. $\frac{1}{\sqrt{1 - x^2}}$

c. $1 - x^2$

d. $\sqrt{x^2 - 1}$

41 [Challenging] • Target: Visualizing a piecewise graph • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Find the cubic polynomial $P(x)$ with leading coefficient 2 whose graph passes through $(0, 6)$ and has zeros $1, -3$ and $1 + i$. But wait — to have real coefficients, if $1 + i$ is a root, so is $1 - i$. That gives degree 4.

a. $P(x) = 2x^3 - 6$

b. $P(x) = 2(x - 1)(x + 3)(x - 1 - i)$

c. $P(x) = 2(x - 1)^2(x + 3)$

d. This is impossible for a cubic with real coefficients (need degree ≥4 for those zeros)

42 [Challenging] • Target: Implicit relations and testing for functions • Cross-Lesson: 1.03 Rates of Change & Concavity

Consider the rational function $f(x) = \frac{x^2 - 9}{x - 3}$. What is the value of $f(4)$?

a. 7

b. 1

c. 5

d. 13

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43 [Challenging] • Target: Implicit differentiation-style reasoning (algebraic slope) • Cross-Lesson: 2.01 Exponential Functions & Growth Models

For $f(x) = (x - 4)^3(x + 10)$, describe the behavior of the graph at $x = 4$.

a. crosses the x-axis normally (linear)

b. crosses the x-axis, flattening (S-shaped) near the zero

c. touches the x-axis and turns around (does not cross)

d. the graph has a vertical asymptote there

44 [Challenging] • Target: Modeling with piecewise functions • Cross-Lesson: 2.02 Logarithmic Functions & Properties

The graph of $y = \sin x$ passes through $(0, 0)$. True or False?

a. Only in Quadrant I

b. False

c. Only for $x > 0$

d. True

45 [Challenging] • Target: Piecewise functions with three or more pieces • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Find the x-coordinate of the hole in the graph of $f(x) = \frac{x^2 + x - 6}{x^2 - 7x + 10}$.

a. x=5

b. x=2

c. No hole exists

d. x=-3

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